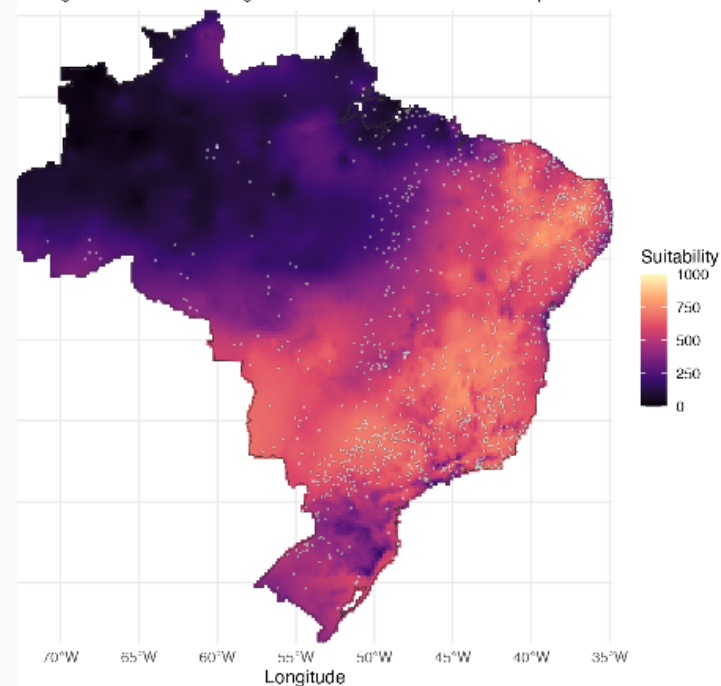


Mapping dengue vector suitability in Brazil

A beginner-friendly species distribution modeling workshop in R

Aedes aegypti + GBIF + WorldClim + biomod2

Predicted environmental suitability for *Aedes aegypti*
Machine learning model for Brazil using GBIF occurrences and bioclimatic predictors



Suitability is not dengue risk. Interpretation requires vector biology, surveillance context, and uncertainty checks.

The public health question starts simple

Where could the environment support the vector?

For dengue, vector ecology is one layer of transmission risk. SDMs help us turn scattered observations into a map of environmental suitability.

Presence records

Where the mosquito has been observed

Environmental predictors

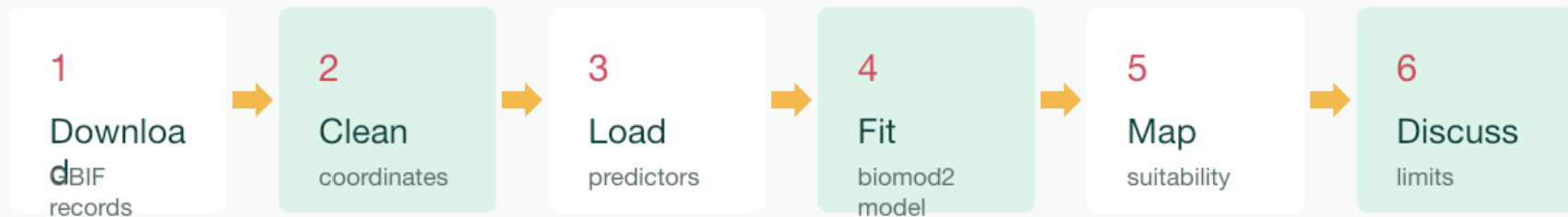
Climate layers across the study area

Suitability map

A hypothesis about where conditions fit

Today's model follows one clear path

The workshop scripts keep each step visible.



The important habit: inspect before you model, and interpret before you act.

GBIF gives us a starting point, not a perfect truth

Open biodiversity data are powerful, but they carry collection bias.

- **Useful**

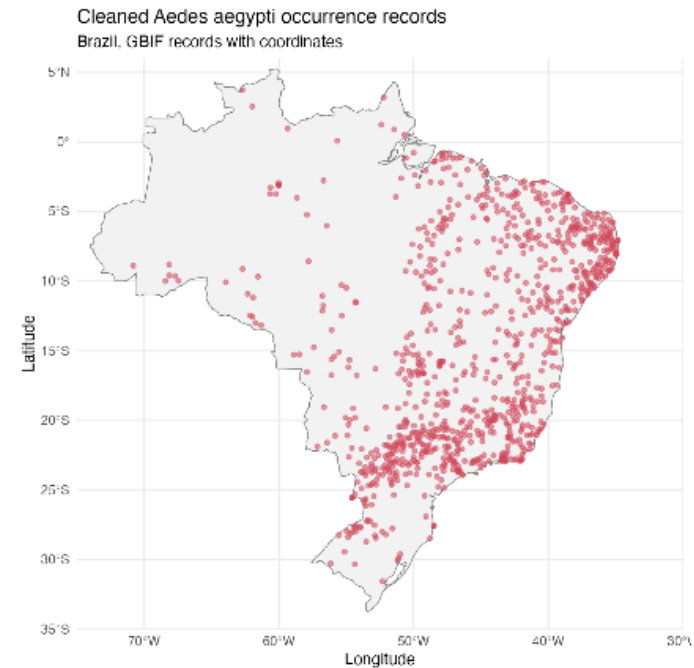
Thousands of public records can be queried from R.

- **Messy**

Coordinates may be duplicated, incomplete, or biased toward sampled places.

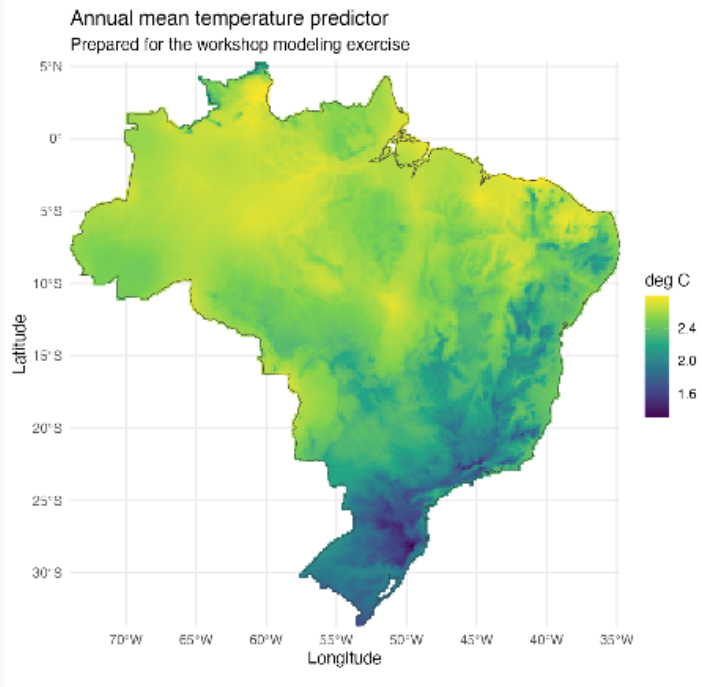
- **Teachable**

Cleaning is part of the model, not a boring prelude.



Predictors describe the stage where the vector could live

We use a small set of climate variables so the first model stays readable.



Annual mean temperature

Temperature seasonality

Annual precipitation

Precipitation seasonality

Simple does not mean final. It means we can see what the workflow is doing.

biomod2 handles the modeling mechanics

Our job is to understand the assumptions we feed into it.

Format presence points + environmental raster layers

Create pseudo-absences for comparison

Fit a simple GLM for the live workshop

Evaluate AUC and TSS as quick diagnostics

Project suitability across Brazil

Live model

GLM

Fast enough
for beginner
laptops

Pseudo-absences are a choice, not a fact

Presence-only data need comparison points, and that choice shapes the model.

What we know

These places have recorded *Aedes aegypti*.

They are presences, not a complete survey of Brazil.

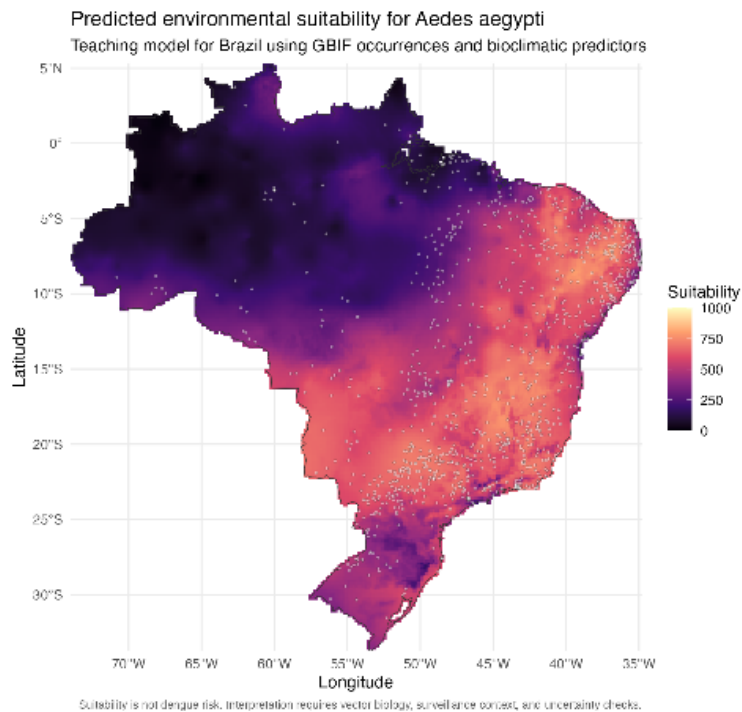
What we compare

Pseudo-absences describe available environments.

Different strategies can change the map.

This is why SDMs are scientific arguments, not map-making buttons.

Read the suitability map as a hypothesis



* environments.

- Where is suitability high?
- Where are occurrence records sparse?
- What could be climate signal?
- What could be sampling bias?

A good SDM result should start a better conversation.

A vector SDM answers only part of the dengue question

That is a strength when we name the boundary clearly.

It can help with

- Vector suitability
- Surveillance prioritization
- Field sampling hypotheses
- Environmental comparison

It does not replace

- Case surveillance
- Vector abundance data
- Local control knowledge
- Transmission modeling

Your job today is not to become an SDM expert

It is to leave with a working map, a reusable R workflow, and better questions.

Run

the scripts line by line

Look

at the data before
modeling

Ask

what the map can and
cannot say

Let's build the model.